Lab Final Project (35 Marks)

Early Detection of Autism Spectrum Disorder (ASD) Using Multi-Modal Machine Learning

In this project, students will design and implement a machine learning-based system for the early detection of Autism Spectrum Disorder (ASD) using multiple types of datasets. Students are encouraged to think beyond a single dataset and develop a multi-model system, where each model works on a different type of data, and the final prediction is generated by combining outputs from all models.

Dataset Requirements

Students may choose from:

- Tabular data

- Image data

- Text data

- Audio data

Datasets can be collected from online repositories.

Rules:

- Minimum: one dataset (pass requirement)

- Recommended: multiple datasets for higher marks

Model Development

Students will:

- Build separate models for each dataset type

- Train and evaluate models independently

-Combine outputs using:

> Confidence averaging

> Weighted averaging

> Voting

> Tournament method

> Any creative approach

System Development

Create a Jupyter Notebook system that:

- Accepts user input
- Runs models
- Combines outputs
- Displays final prediction

Output & Interpretation

- Show ASD / No ASD result
- Provide natural language explanation

Explainability (Optional Bonus)

To get bonus mark one can use one of the following techniques to explain output:

- Feature importance
- LIME / SHAP

Evaluation (35 Marks)

- Understanding of Dataset: 5
- Model (Single): 8
- Multi-dataset: 7
- Combining results: 5

- System for user input and output: 5

- Interpretation of result: 3

- Code: 2

- Bonus: +2

Notes

- Add comments and explanations in notebook
- Creativity in model combination is encouraged
- Simpler working solutions are better than complex incomplete ones
- Focus on understanding rather than complexity
- No copy from others